

Git for Developers

Complete Course

Version Control from First Commit to Advanced Workflows

Course Overview

What You Will Learn

- Why version control matters and the problems Git solves
- Core Git concepts: the three-tree architecture, staging, commits
- Branching strategies and merging techniques
- Remote collaboration with GitHub, GitLab, and Bitbucket

Advanced Topics

- Rebasing, cherry-picking, stashing, and tagging
- Common team workflows: Gitflow, trunk-based development
- Optimization, commit conventions, and best practices
- Git internals: object model, hooks, and plumbing commands
- Enterprise patterns: monorepos, LFS, and signed commits

Prerequisites

Topic	Expected level
Command line basics	Comfortable with a terminal
Text editing	Any editor (VS Code recommended)
Networking basics	Helpful but not required

No prior Git experience required. We start from the very beginning.

Modules

#	Module	Key Topics
1	Introduction & Basics	Version control, installation, setup
2	Core Concepts	Staging area, commits, history, undo
3	Branching & Merging	Branches, fast-forward, conflicts
4	Remote Collaboration	GitHub, push/pull, forking
5	Advanced Features	Rebase, stash, tags, submodules
6	Workflows	Gitflow, trunk-based, feature branches
7	Best Practices	Commit messages, naming, performance
8	Debugging & History	bisect, reflog, clean
9	Git Internals	Objects, hooks, plumbing

Ready?

Let's start from the beginning.